

Bhavani Ayappann

Senior Frontend Engineer

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Fremont, California • EAD (Authorized to Work in U.S. — No sponsorship required)

PROFESSIONAL SUMMARY

Senior Frontend Engineer with 13+ years of experience architecting scalable, high-performance web applications for product companies across India, Singapore, and the U.S. Deep expertise in React, TypeScript, Next.js, geospatial visualization (D3.js, MapLibre GL, WebGL), and modern data-intensive UIs. Proven track record shipping intuitive, performant user experiences. Managed significant personal milestones (maternity, relocation to California) while maintaining hands-on engineering practice and staying current with modern full-stack ecosystem advances. Currently building AI-native SaaS tools with production-grade discipline and mastering LLM integration patterns. Strong background in architectural decision-making and cross-functional leadership.

CORE TECHNICAL SKILLS

Frontend: React.js, Next.js, TypeScript, JavaScript (ES6+), Redux, Zustand, React Server Components, Custom Hooks, Responsive UI

AI & LLMs: OpenAI API, Claude API, Vercel AI SDK, function calling, streaming APIs, prompt engineering

Visualization & GIS: D3.js, MapLibre GL, react-map-gl, Leaflet.js, WebGL, Three.js, Highcharts, ECharts

State Management & Architecture: Zustand, Redux, custom hooks, URL state synchronization, data fetching patterns

Backend & Data: Supabase (PostgreSQL, Auth, Storage, real-time), Node.js, REST APIs, SQL

Tooling & DevOps: Git, Webpack, npm, Vercel, ESLint, TypeScript strict mode, performance profiling

Other: Figma, Material-UI, Ant Design, Tailwind CSS, SASS, Angular, ArcGIS

SABBATICAL & ADVANCED ENGINEERING PROJECTS

Maternity, Relocation & Product Development

Fremont, California • Jan 2024 – Present

Took time away from traditional employment to manage significant personal milestones—maternity leave and relocation. Once settled, pivoted focus to hands-on AI-native product development while maintaining engineering velocity. Built portfolio-quality projects demonstrating production-grade discipline, product thinking, and emerging LLM integration patterns. Stayed current with modern full-stack ecosystem while managing family priorities.

Neighbourly – AI-Native Neighborhood Analysis SaaS

neighbourly.live | Next.js 14 + React 18 + MapLibre GL + D3.js + OpenAI API

- Architected and shipped full-stack AI-native SaaS from Figma design through Vercel production; owned architectural decisions across state management (Zustand + custom hooks), API strategy, Supabase caching layers, and React component hierarchy
- Engineered MapLibre GL choropleth map with D3 mini-charts, county-level hover interactions, and school/commute overlays; achieved sub-200ms render time on 3000+ Census tracts by optimizing layer composition, using WebWorkers for data processing, and implementing viewport-aware tooltip positioning
- Designed two-tier caching strategy (Supabase RLS + in-memory Zustand cache) that reduced OpenAI API calls by 65% through prompt context memoization, batched requests during geographic pan/zoom, and smart cache invalidation on user actions
- Integrated LLMs across three distinct patterns: (1) grounded chatbot with injected Census demographics + Reddit sentiment, (2) real-time sentiment scoring on news feeds using Claude API with structured JSON outputs, (3) neighborhood match explanations with transparent scoring logic—all with graceful fallbacks when APIs unavailable

- Managed complete product ownership: user research → wireframing in Figma → technical specification → GitHub-based feature branches → Vercel deployment; documented architectural decisions (state flows, API contracts, performance budgets) in shared decision log
- Mastered modern AI-native patterns: streaming API responses with React state updates, function calling workflows for structured data extraction, handling token limits across multi-turn conversations, and synchronizing real-time data with LLM context windows

PROFESSIONAL EXPERIENCE

Senior Frontend Developer | Chesslang — Bootstrapped EdTech Chess Platform

Equity-Stake Contributor

2023 – 2025

- Owned architecture and shipping decisions end-to-end for high-impact features (certificate builder, tournament management UI) that drove measurable engagement and retention
- Built interactive chess board components with real-time piece validation, lesson modules with progress tracking, and tournament brackets; collaborated closely with founding team on product roadmap and performance optimization

Senior Frontend Engineer

ObjectEdge

Sep 2021 – Jun 2022

- Built responsive interfaces for Oracle Commerce and commercetools platforms using React and TypeScript; delivered subscription management and CPQ (Configure-Price-Quote) systems
- Collaborated in agile teams to ship scalable, enterprise-grade e-commerce solutions for Fortune 500 clients

Lead Solution Engineer

Zinier Asia Pte Ltd (Series C) — Singapore

Sep 2019 – Dec 2020

- Led technical architecture for field service automation platform; delivered executive dashboards using JIRA and DOMO for real-time tracking efficiency
- Coordinated globally across San Mateo, Mexico, India, and Singapore for 24/7 delivery cadence

Senior Web Developer – R&D

NCS Pte Ltd — Singapore

Apr 2016 – Sep 2019

- Architected Smart-City platform for government clients using React, Redux, and GIS tools; built real-time 3D dashboards and multi-layer mapping interfaces with WebGL for enterprise data visualization
- Migrated legacy jQuery applications to React, improving performance and maintainability across the platform

Web Developer

Cognizant Interactive — Chennai

Dec 2009 – Apr 2016

- Shipped 15+ enterprise applications across industries using JavaScript, HTML/CSS, and AngularJS; led UI refactors and CMS integrations (dotCMS, Velocity) to improve maintainability and scalability

EDUCATION

Bachelor of Engineering in Computer Science

College of Engineering, Guindy (Anna University) • GPA: 8.1/10

ADDITIONAL

Work Authorization: EAD (Authorized to Work in U.S. — No sponsorship required)

Languages: English (Fluent), Tamil (Native)

Portfolio: neighbourly.live